

30/09/26

Signal Processing

• Recap: DTFT theorems: Parseval's theorem, shift in the freq. domain, windowing theorem

• Rensit modulation

• Z-transform examples

$$- x[n] = a^n u[n]$$

} comparison

$$- x[n] = -a^n \cdot u[-n-1]$$

$$- x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{3}\right)^n u[n]$$

$$- x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-n-1]$$

} comparison, DTFT

• ROC properties: zero-centered ring or disk, DTFT of ROC includes unit circle, no poles,

finite duration, right-sided, left-sided, two-sided, connected regions

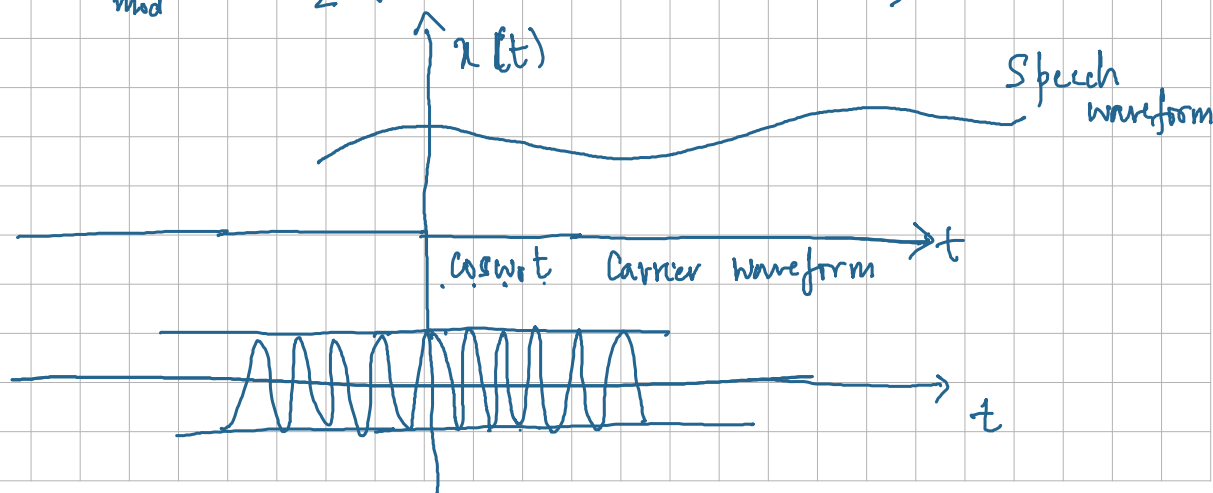
• Rensit modulation result:

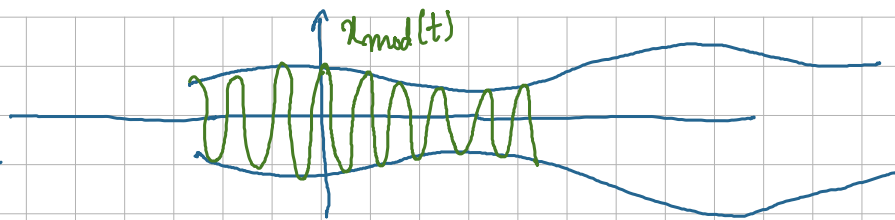
$$x[n] \leftrightarrow X(e^{j\omega})$$

$$\underbrace{e^{j\omega_0 n}} \underbrace{x[n] \leftrightarrow X(e^{j(\omega-\omega_0)})}$$

Example: $x_{mod}[n] = \cos \omega_0 n \cdot x[n] = x[n] \cos \omega_0 n$

$$X_{mod}(e^{j\omega}) = \frac{1}{2} (X(e^{j(\omega-\omega_0)}) + X(e^{j(\omega+\omega_0)}))$$





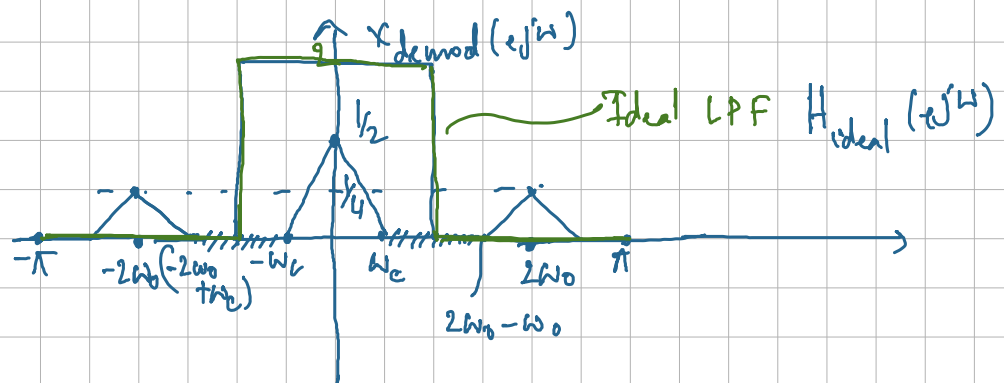
$$x_{\text{demod}}[n] = \cos \omega_c n \cdot x_{\text{mod}}[n]$$

$$= \cos \omega_c n \cdot (\cos \omega_0 n \cdot x[n])$$

$$= \cos^2 \omega_c n \cdot x[n]$$

$$= \left(1 + \frac{\cos 2\omega_0 n}{2}\right) x[n]$$

$$\Rightarrow X_{\text{demod}}(e^{j\omega}) = \frac{X(e^{j\omega})}{2} + \frac{1}{4} \left(X(e^{j(\omega-2\omega_0)}) + X(e^{j(\omega+2\omega_0)}) \right)$$



This is a sample AM system

• Z-Transform

$$\bullet \quad x[n] = a^n \cdot u[n]$$

$$\Rightarrow X(z) = \underbrace{\frac{1}{1-az^{-1}}}_{\text{region of convergence}} ; \underbrace{|z| > |a|}_{\text{region of convergence}} \quad - (1)$$

$$\bullet \quad x[n] = -a^n u[-n-1]$$

$$X(z) = \frac{1}{1-az^{-1}} ; |z| < |a| \quad - (2)$$

While (1) and (2) contain the same expression for $X(z)$, they correspond to different $x[n]$ as differentiated by the ROC.

$$\bullet x[n] = \underbrace{\left(\frac{1}{2}\right)^n \cdot u[n]} + \underbrace{\left(-\frac{1}{3}\right)^n \cdot u[n]}$$

$$= \frac{1}{1 - \frac{z^{-1}}{2}} + \frac{1}{1 + \frac{z^{-1}}{3}} ; \quad \mathcal{R}_1 \cap \mathcal{R}_2$$

$$\mathcal{R}_1 = \{z \mid |z| > \frac{1}{2}\} \checkmark \checkmark$$

$$\mathcal{R}_2 = \{z \mid |z| > \frac{1}{3}\}$$